Madhav Umesh Kamble

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Education

Indian Institute of Information Technology Allahabad — Prayagraj, Uttar Pradesh
M.Tech in Information Technology (Data Engineering) — *Aug. 2025 – July 2027*
- CGPA: 7.89 / 10
- Research: *Missing Data Imputation via Genetic Algorithm*
- Research: *Enhancing Genetic Algorithm with Explainable AI for Last-Mile Routing*

SKN Sinhgad College of Engineering — Korti, Pandharpur
B.Tech in Computer Science and Engineering — *Aug. 2021 – May 2025*
- CGPA: 9.31 / 10

Experience

Leadsoft IT Solutions — Virtual
Full Stack Web Development Intern — *Aug. 2023 – Oct. 2023*
- Built a Gate Pass Management mobile app (React Native + MERN stack) supporting digital pass requests, multi-level approvals, and real-time tracking.
- Developed RESTful APIs in Node.js/Express with MongoDB for pass lifecycle management and real-time status updates.

Projects

Real-Time Ride-Sharing Analytics Platform — Jan. 2026 – May 2026
Python, Kafka, PySpark, Delta Lake, Airflow, XGBoost, MLflow, Redis, Streamlit, Docker
- Built an end-to-end streaming platform simulating Uber/Ola internals: Kafka ingestion into PySpark Structured Streaming, Delta Lake medallion architecture, ML-powered surge pricing, and a live Streamlit dashboard.
- Designed Bronze-Silver-Gold pipeline processing 8 events/sec across 3 Kafka topics; served XGBoost surge model via MLflow registry and PySpark Pandas UDF.
- Reduced dashboard latency from approx. 10s to under 1ms with Redis cache; orchestrated 4 Airflow DAGs (hourly refresh, daily retrain, quality checks, cache warmup) across 10 Docker services.

Traffic Demand Prediction — GitHub — Apr. 2026 – May 2026
Python, LightGBM, XGBoost, CatBoost, Pandas, NumPy
- Built an ML pipeline to predict normalized traffic demand across 1,200+ geographic zones, achieving R2 score of 91.8% – ranked Top 1.5% in a competitive hackathon.
- Engineered a 3-model GBT ensemble with pseudo-labeling, residual modeling, and geohash-based spatial features to overcome train/test distribution shift.

MIGA: Missing Data Imputation via Genetic Algorithm — GitHub — Jan. 2026 – Present
Python, NumPy, SciPy, scikit-learn, Pandas, Matplotlib, Genetic Algorithms
- Reimplemented MIGA (Figueroa-García et al., *Information Sciences* 2023) as the first open-source Python implementation; benchmarked against MICE, KNN, and mean imputation across 7 UCI datasets with Wilcoxon significance tests.
- Extended with Ledoit-Wolf shrinkage covariance, adaptive mutation scheduling, MNAR evaluation, variance preservation analysis, and a kurtosis-augmented fitness function (Fr+).

Achievements

LeetCode: Rating 1669
CodeChef: Rating 1441
Runner-Up – Sinhgad Hackathon 2023
Runner-Up – Sinhgad Hackathon 2022

Certifications

0 to 100 Full-Stack Cohort – 100xDevs (2024)
Cyber Threat Intelligence 101 – arcX (2024)
Getting Started with Enterprise Data Science – IBM (2024)

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML, CSS
Data & ML: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, Matplotlib, Seaborn, Hadoop
Data Engineering: Apache Kafka, PySpark, Delta Lake, Apache Airflow, MLflow, Redis
Web Frameworks: React.js, Next.js, Node.js, Express.js, Tailwind CSS
Databases: MongoDB, MySQL, PostgreSQL
Tools & Platforms: Git, GitHub, Docker, Linux, VS Code
CS Fundamentals: DSA, OS, CN, OOP, DBMS, ML, Big Data, Data Visualisation